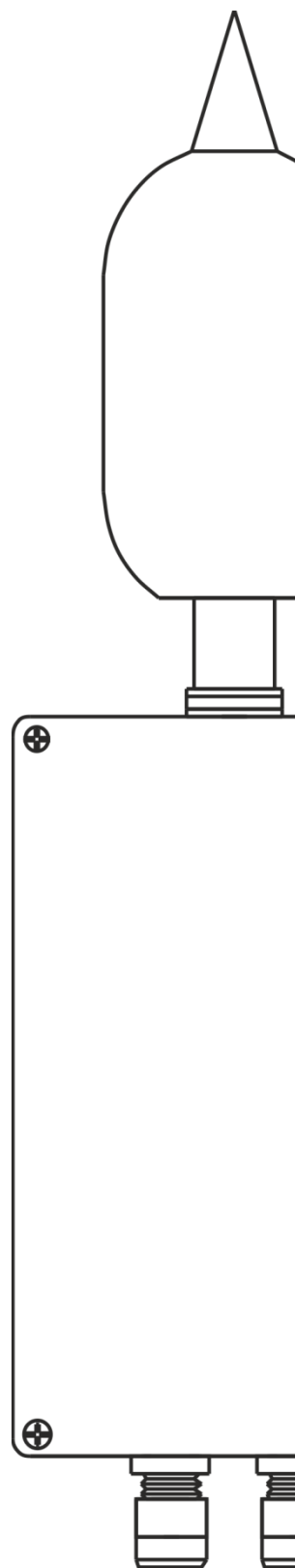


## TA150

### EXTERNAL OUTPUT PROTOCOLS

PR\_TA150output\_v0002\_20250415\_EN



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The TA150 sensor has several external output hardware modules that are used for communication. These modules are RS152\* (RS-232 serial), RS154\* (RS-485 serial) and CL150\* (current loop). The user can purchase the one that best suits his needs.

Depending on the hardware module available, the sensor can communicate using certain protocols. To select the protocol and configure it, it will be done through the web server as shown below.

**IMPORTANT:** This manual mentions modules that are not included in the basic configuration of the equipment, which are optional and must be purchased additionally.

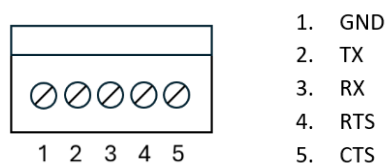
These modules will be indicated with an (\*).

Only one additional hardware module can be added (see chapter 6 of the User Manual).

It is important to bear in mind that, once the module has been installed, it cannot be modified.

**NOTE:** Throughout this manual, transmission refers to sending data from the TA150 sensor to the PC.

Protocols for RS232 can only be implemented if the RS152\* hardware module has been purchased.



**Connector terminals**

The information displayed by the web server is:

- The installed output interface
- The selected protocol
- And the protocol configuration

To modify the protocol:

- Press the  button.
- Select the desired option from the drop-down menu.
- Finally click  to save the change or click  to discard it.

The available protocols are:

1. Disabled
2. CESVA
3. Modbus

## 2.1 Disabled

---

No parameters will have to be configured as the option will be disabled.

## 2.2 CESVA

---

### 2.2.1 Configuration

Through the web server, the following parameters must be configured:

- |                         |                               |
|-------------------------|-------------------------------|
| • Averaging time (TLeq) | From 1 to 3600 seconds        |
| • Flow control          | None, CTS, RTS or RTS and CTS |
| • Baud rate             | maximum 115200 bps            |
| • Parity                | None, even, odd               |
| • Number of stop bits   | 1, 1.5 or 2                   |

Finally press  .

### 2.2.2 Protocol

- **FROM MASTER TO TA150**

MASTER commands to TA150 must always end with a carriage return CR (13).

- |       |                                                              |
|-------|--------------------------------------------------------------|
| ▪ "M" | TRANSMISSION ON. It starts packet transmission every second. |
| ▪ "m" | TRANSMISSION OFF. End transmission.                          |

- **FROM TA150 TO MASTER**

*Every second the next packet is transmitted:*

- STX (02)
- M
- t of T (4 characters: SSSS)
- L A T
- L C T
- L Z T
- L A I T

- L A F max T
- L A S max T
- L A I max T
- L C peak T
- L Z peak T
- ETX (03)
- CR (13)
- LF (10)

Levels are sent with 5 characters and the following format:

- Saturation: '-' no, 'o' overload, 'u' underrange, 'b' overload and underrange.
- Level x 10: 4 digits.

### 2.2.2.1 EXAMPLE Consultation of measurements with Tera Term

M0035u0286-0364-0392-0356-0479-0431-0406-0301-0320

Where

M– Second data package

0035 – 35 seconds elapsed from T

u – underrange

0286 – level of the measured LAT function: 64,6 dB

(0286 in hexadecimal is 646 in decimal:  $2 \times 256 + 8 \times 16 + 6 \times 1$ )

'-' – no overload or underrange

0364 – level of the measured LCT function: 86,8 dB

'-' – no overload or underrange

0392 – level of the measured LZT function: 91,4 dB

'-' – no overload or underrange

0356 – level of the measured LAIT function: 85,4 dB

'-' – no overload or underrange

0479 – level of the measured LAFmaxT function: 114,5 dB

'-' – no overload or underrange

0431 – level of the measured LASmaxT function: 107,3 dB

'-' – no overload or underrange

0406 – level of the measured LAImaxT function: 103,0 dB

'-' – no overload or underrange

0301 – level of the measured LCPeakT function: 76,9 dB

'-' – no overload or underrange

0320 – level of the measured LZPeakT function: 80,0 dB

## 2.3 MODBUS

---

### 2.3.1 Configuration

Through the web server, the following parameters must be configured:

- Averaging time (TLeq) From 1 to 3600 seconds
- Flow control None
- Baud rate maximum 115200 bps
- Parity None, even, odd
- Stop bits 1, 1.5 or 2
- Address (MODBUS Address) 001

Finally press .

### 2.3.2 Protocol

- **REGISTERS**

The measurements are in the following *Holding Registers*:

| Register | Parameter                                                   | Type of data      | Value     | Range         |
|----------|-------------------------------------------------------------|-------------------|-----------|---------------|
| 0000     | Elapsed time of the interval T, in seconds                  | 16 bits, unsigned | 1 to 3600 | 1 to 3600 s   |
| 0001     | L <sub>AT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0002     | <i>Overload/Under-range</i> of the L <sub>AT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0003     | L <sub>CT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0004     | <i>Overload/Under-range</i> of the L <sub>CT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0005     | L <sub>ZT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |

|      |                                                                 |                   |           |               |
|------|-----------------------------------------------------------------|-------------------|-----------|---------------|
| 0006 | <i>Overload/Under-range</i> of the L <sub>ZT</sub> function     | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0007 | L <sub>AIT</sub> function level, in dBs                         | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0008 | <i>Overload/Under-range</i> of the L <sub>AIT</sub> function    | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0009 | L <sub>AFmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 000A | <i>Overload/Under-range</i> of the L <sub>AFmaxT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000B | L <sub>ASmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 000C | <i>Overload/Under-range</i> of the L <sub>ASmaxT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000D | L <sub>AlmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 000E | <i>Overload/Under-range</i> of the L <sub>AlmaxT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000F | L <sub>CpeakT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0010 | <i>Overload/Under-range</i> of the L <sub>CpeakT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0011 | L <sub>ZpeakT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0012 | <i>Overload/Under-range</i> of the L <sub>ZpeakT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |

The meaning of the Overload/Under-range registers is as follows:

- 0 – none
- 1 – *under-range*
- 2 – *overload*
- 3 – *under-range and overload*



### 2.3.2.1 MEASUREMENT CONSULTATION

---

Measurements can be queried with the 0x03 function, reading of *Holding Registers*.  
If the address of TA150 is 01, the frame to query the first three registers is:

01 03 00 00 00 03 05 CB

Where:

01 – TA150 address  
03 – function: reading of *Holding Registers*  
0000 – first register to be read  
0003 – number of registers to be read  
05CB – CRC (CB05)

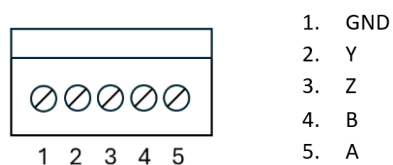
And a possible answer:

01 03 06 00 13 02 41 00 00 F5 1A

Where:

01 – TA150 address  
03 – function: reading of *Holding Registers*  
06 – number of bytes  
0013 – elapsed time of the interval T: 19 seconds (0013 in hexadecimal is 19 in decimal:  $1 \times 16 + 3$ )  
0241 – level of the measured function: 57,7 dB (0241 in hexadecimal is 577 in decimal:  $2 \times 256 + 4 \times 16 + 1$ )  
0000 – no *under-range* or *overload*  
F51A – CRC (1AF5)

Protocols for RS485 can only be implemented if the RS154\* hardware module has been purchased.



**Connector terminals**

The information displayed by the web server is:

- The installed output interface
- The selected protocol
- And the protocol configuration

To modify the protocol:

- Press the  button.
- Select the desired option from the drop-down menu.
- Finally click  to save the change or click  to discard it.

The available protocols are:

1. Disabled
2. CESVA
3. Modbus
4. Modbus (2-Wire, 4-Wire)

### 3.1 Disabled

---

No parameters will have to be configured as the option will be disabled.

### 3.2 CESVA

---

#### 3.2.1 Configuration

Through the web server, the following parameters must be configured:

- |                                               |                        |
|-----------------------------------------------|------------------------|
| • Averaging time (TLeq)                       | From 1 to 3600 seconds |
| • RXR, Receiver Termination Resistor (A/B)    | Falso o Verdadero      |
| • TXR, Transmitter Termination Resistor (Y/Z) | Falso o Verdadero      |
| • Baud rate                                   | maximum 115200 bps     |
| • Parity                                      | None, even, odd        |
| • Number of stop bits                         | 1, 1.5 or 2            |

Finally press .

The transmission is automatic.  
(There cannot be more slaves)

- Transmission Y/Z
- Reception A/B (optional)

#### 3.2.2 Protocol

- **FROM MASTER TO TA150**

Los comandos del MASTER al TA150 han de terminar siempre con un retorno de carro CR (13).

- |       |                                                              |
|-------|--------------------------------------------------------------|
| ▪ "M" | TRANSMISSION ON. It starts packet transmission every second. |
| ▪ "m" | TRANSMISSION OFF. End transmission.                          |

- **FROM TA150 TO MASTER**

*Every second the next packet is transmitted:*

- STX (02)
- M
- t of T (4 characters: SSSS)
- L A T
- L C T
- L Z T
- L A I T
- L A F max T
- L A S max T
- L A I max T
- L C peak T
- L Z peak T
- ETX (03)
- CR (13)
- LF (10)

Levels are sent with 5 characters and the following format:

- Saturation: '-' no, 'o' overload, 'u' underrange, 'b' overload and underrange.
- Level x 10: 4 digits.

### 3.2.2.1 EXAMPLE Consultation of measurements with Tera Term

M0035u0286-0364-0392-0356-0479-0431-0406-0301-0320

Where

M– Second data package

0035 – 35 seconds elapsed from T

u – underrange

0286 – level of the measured LAT function: 64,6 dB

(0286 in hexadecimal is 646 in decimal:  $2 \times 256 + 8 \times 16 + 6 \times 1$ )

'-' – no overload or underrange

0364 – level of the measured LCT function: 86,8 dB

'-' – no overload or underrange

0392 – level of the measured LZT function: 91,4 dB

'-' – no overload or underrange

0356 – level of the measured LAIT function: 85,4 dB

‘-‘ – no overload or underrange  
 0479 – level of the measured LAFmaxT function: 114,5 dB  
 ‘-‘ – no overload or underrange  
 0431 – level of the measured LASmaxT function: 107,3 dB  
 ‘-‘ – no overload or underrange  
 0406 – level of the measured LAlmaxT function: 103,0 dB  
 ‘-‘ – no overload or underrange  
 0301 – level of the measured LCPeakT function: 76,9 dB  
 ‘-‘ – no overload or underrange  
 0320 – level of the measured LZPeakT function: 80,0 dB

### 3.3 MODBUS (2-Wire)

---

#### 3.3.1 Configuration

Through the web server, the following parameters must be configured:

- |                                               |                        |
|-----------------------------------------------|------------------------|
| • Averaging time (TLeq)                       | From 1 to 3600 seconds |
| • RXR, Receiver Termination Resistor (A/B)    | Falso o Verdadero      |
| • TXR, Transmitter Termination Resistor (Y/Z) | Falso o Verdadero      |
| • Baud rate                                   | maximum 115200 bps     |
| • Parity                                      | None, even, odd        |
| • Stop bits                                   | 1, 1.5 or 2            |
| • Address (MODBUS Address)                    | 001                    |

Finally press  .

If desired, more slaves may be present.

- Transmission Y/Z
- Reception Y/Z

### 3.3.2 Protocol

- **REGISTERS**

The measurements are in the following *Holding Registers*:

| Register | Parameter                                                       | Type of data      | Value     | Range         |
|----------|-----------------------------------------------------------------|-------------------|-----------|---------------|
| 0000     | Elapsed time of the interval T, in seconds                      | 16 bits, unsigned | 1 to 3600 | 1 to 3600 s   |
| 0001     | L <sub>AT</sub> function level, in dBs                          | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0002     | <i>Overload/Under-range</i> of the L <sub>AT</sub> function     | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0003     | L <sub>CT</sub> function level, in dBs                          | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0004     | <i>Overload/Under-range</i> of the L <sub>CT</sub> function     | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0005     | L <sub>ZT</sub> function level, in dBs                          | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0006     | <i>Overload/Under-range</i> of the L <sub>ZT</sub> function     | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0007     | L <sub>AIT</sub> function level, in dBs                         | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0008     | <i>Overload/Under-range</i> of the L <sub>AIT</sub> function    | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0009     | L <sub>AFmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 000A     | <i>Overload/Under-range</i> of the L <sub>AFmaxT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000B     | L <sub>ASmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 000C     | <i>Overload/Under-range</i> of the L <sub>ASmaxT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000D     | L <sub>AlmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |

|      |                                                           |                   |           |               |
|------|-----------------------------------------------------------|-------------------|-----------|---------------|
| 000E | <i>Overload/Under-range</i> of the $L_{A\max T}$ function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000F | $L_{CpeakT}$ function level, in dBs                       | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0010 | <i>Overload/Under-range</i> of the $L_{CpeakT}$ function  | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0011 | $L_{ZpeakT}$ function level, in dBs                       | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0012 | <i>Overload/Under-range</i> of the $L_{ZpeakT}$ function  | 16 bits, unsigned | 0 to 3    | 0 to 3        |

The meaning of the *Overload/Under-range* registers is as follows:

- 0 – none
- 1 – *under-range*
- 2 – *overload*
- 3 – *under-range and overload*

### 3.3.2.1 MEASUREMENT CONSULTATION

---

Measurements can be queried with the 0x03 function, reading of *Holding Registers*.  
If the address of TA150 is 01, the frame to query the first three registers is:

01 03 00 00 00 03 05 CB

Where:

- 01 – TA150 address
- 03 – function: reading of *Holding Registers*
- 0000 – first register to be read
- 0003 – number of registers to be read
- 05CB – CRC (CB05)

And a possible answer:

01 03 06 00 13 02 41 00 00 F5 1A

Where:

01 – TA150 address

03 – function: reading of *Holding Registers*

06 – number of bytes

0013 – elapsed time of the interval T: 19 seconds (0013 in hexadecimal is 19 in decimal:  $1 \times 16 + 3$ )

0241 – level of the measured function: 57,7 dB (0241 in hexadecimal is 577 in decimal:  $2 \times 256 + 4 \times 16 + 1$ )

0000 – no *under-range* or *overload*

F51A – CRC (1AF5)

## 3.4 MODBUS (4-Wire)

---

### 3.4.1 Configuration

Through the web server, the following parameters must be configured:

- |                                               |                        |
|-----------------------------------------------|------------------------|
| • Averaging time (TLeq)                       | From 1 to 3600 seconds |
| • RXR, Receiver Termination Resistor (A/B)    | Falso o Verdadero      |
| • TXR, Transmitter Termination Resistor (Y/Z) | Falso o Verdadero      |
| • Baud rate                                   | maximum 115200 bps     |
| • Parity                                      | None, even, odd        |
| • Stop bits                                   | 1, 1.5 or 2            |
| • Address (MODBUS Address)                    | 001                    |

Finally press .

If desired, more slaves may be present.

- Transmission Y/Z
- Reception Y/Z



## 3.4.2 Protocol

- **REGISTERS**

The measurements are in the following *Holding Registers*:

| Register | Parameter                                                       | Type of data      | Value     | Range         |
|----------|-----------------------------------------------------------------|-------------------|-----------|---------------|
| 0000     | Elapsed time of the interval T, in seconds                      | 16 bits, unsigned | 1 to 3600 | 1 to 3600 s   |
| 0001     | L <sub>AT</sub> function level, in dBs                          | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0002     | <i>Overload/Under-range</i> of the L <sub>AT</sub> function     | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0003     | L <sub>CT</sub> function level, in dBs                          | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0004     | <i>Overload/Under-range</i> of the L <sub>CT</sub> function     | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0005     | L <sub>ZT</sub> function level, in dBs                          | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0006     | <i>Overload/Under-range</i> of the L <sub>ZT</sub> function     | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0007     | L <sub>AIT</sub> function level, in dBs                         | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0008     | <i>Overload/Under-range</i> of the L <sub>AIT</sub> function    | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0009     | L <sub>AFmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 000A     | <i>Overload/Under-range</i> of the L <sub>AFmaxT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000B     | L <sub>ASmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 000C     | <i>Overload/Under-range</i> of the L <sub>ASmaxT</sub> function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000D     | L <sub>AlmaxT</sub> function level, in dBs                      | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |

|      |                                                           |                   |           |               |
|------|-----------------------------------------------------------|-------------------|-----------|---------------|
| 000E | <i>Overload/Under-range</i> of the $L_{A\max T}$ function | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 000F | $L_{CpeakT}$ function level, in dBs                       | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0010 | <i>Overload/Under-range</i> of the $L_{CpeakT}$ function  | 16 bits, unsigned | 0 to 3    | 0 to 3        |
| 0011 | $L_{ZpeakT}$ function level, in dBs                       | 16 bits, unsigned | 0 to 1600 | 0 to 160,0 dB |
| 0012 | <i>Overload/Under-range</i> of the $L_{ZpeakT}$ function  | 16 bits, unsigned | 0 to 3    | 0 to 3        |

The meaning of the Overload/Under-range registers is as follows:

- 0 – none
- 1 – *under-range*
- 2 – *overload*
- 3 – *under-range and overload*

### 3.4.2.1 MEASUREMENT CONSULTATION

---

Measurements can be queried with the 0x03 function, reading of *Holding Registers*.

If the address of TA150 is 01, the frame to query the first three registers is:

01 03 00 00 00 03 05 CB

Where:

- 01 – TA150 address
- 03 – function: reading of *Holding Registers*
- 0000 – first register to be read
- 0003 – number of registers to be read
- 05CB – CRC (CB05)

And a possible answer:

01 03 06 00 13 02 41 00 00 F5 1A

Where:

01 – TA150 address

03 – function: reading of *Holding Registers*

06 – number of bytes

0013 – elapsed time of the interval T: 19 seconds (0013 in hexadecimal is 19 in decimal:  $1 \times 16 + 3$ )

0241 – level of the measured function: 57,7 dB (0241 in hexadecimal is 577 in decimal:  $2 \times 256 + 4 \times 16 + 1$ )

0000 – no *under-range* or *overload*

F51A – CRC (1AF5)

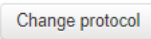
# CURRENT LOOP | 4

Protocols for 4-20 mA can only be implemented if the CL150\* hardware module has been purchased.

The information displayed by the web server is:

- The installed output interface
- The selected protocol
- And the protocol configuration

To modify the protocol:

- Press the  button.
- Select the desired option from the drop-down menu.
- Finally click  to save the change or click  to discard it.

The available protocols are:

1. Disabled
2. CESVA

## 4.1 Disabled

---

No parameters will have to be configured as the option will be disabled.

## 4.2 CESVA

---

### 4.2.1 Configuration

Through the web server, the following parameters must be configured:

- Averaging time (TLeq) From 1 to 3600 seconds
- Function  $L_{AT}$ ,  $L_{CT}$ ,  $L_{ZT}$ ,  $L_{AIT}$ ,  $L_{AFmaxT}$ ,  $L_{ASmaxT}$ ,  $L_{AlmaxT}$ ,  $L_{CpeakT}$  or  $L_{ZpeakT}$

Finally press  .

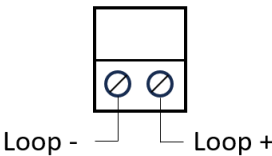
### 4.2.2 Technical characteristics

- Power supply 8 a 36 VDC
- Output 4 to 20 mA / 27.4 to 137.0 dB

$$I_{out} = \left( (L_{AT}) * \frac{20}{137} \right)$$

Note 1:  $I_{out}$  is the output current in mA  
Note 2:  $L_{AT}$  is the measured level in dBA  
Note 3: The measuring range is from 27,4 to 137,0 dBA  
Note 4: T is the programmed averaging time. The output current is updated every T<sup>(\*)</sup>.  
Note 5: If an overload occurs, then  $I_{out} = 24,0$  mA

- Maximum error  $\pm 4 \mu A$
- Connections With polarity



### 4.3 Brief description

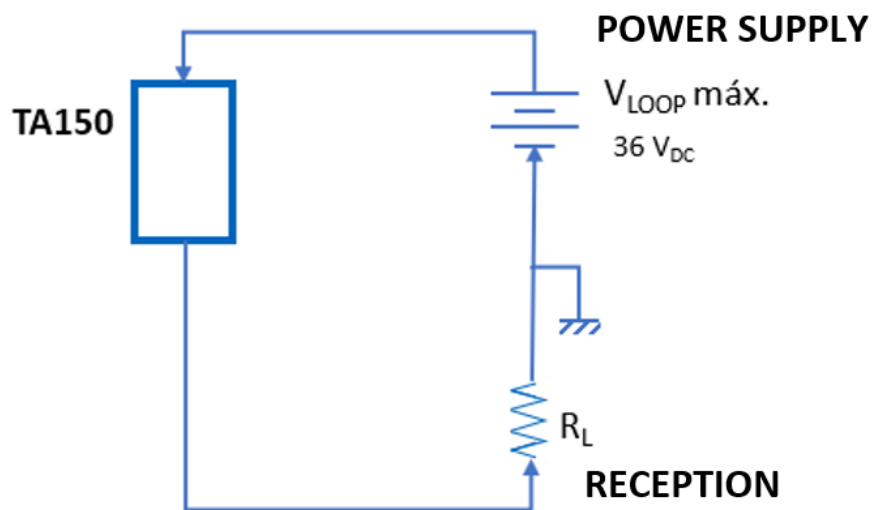
---

A typical 4-20mA current-loop circuit is made up by next elements: a sensor (TA120); a power supply for the loop; and a receiver (to read the information). These elements are connected in a closed, series circuit, loop.

The sensor output transmits the proportional 4-20mA dc-current that circulates within the closed series-loop. The 4 mA represents the sensor's lowest level output, and 20 mA representing the sensor's highest level output.

The receiver, normally a data acquisition system, and converts the 4-20mA current back into a voltage which can be further processed and/or displayed.

Current loops are ideal for data transmission because of their inherent insensitivity to electrical noise. In a 4-20 mA current loop, all the current flows through all components.



**Fig.1** Current loop (4 – 20 mA)





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